Vietnamese Fashion E-Commerce:
Revenue Forecasting & Business Intelligence

Datathon 2026 — The Gridbreaker Technical Report

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https://github.com/vinhnguyen161006/Datathon2026-Vietnamese-ECommerce-Forecasting

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Abstract

We present a complete analytics solution for a Vietnamese fashion e-commerce platform (2012–2022, 16.43B VND total revenue, 646,945 orders, 121,930 customers). **Part 2 (EDA):** Six analytical notebooks reach Prescriptive level across customer lifecycle, promotions, returns, inventory, web traffic, and reviews, identifying over 300M VND in quantified annual opportunities. **Part 3 (Forecasting):** A LightGBM model with 66 engineered features—including five Facebook Prophet seasonality components as adaptive inputs—achieves **MAE 808,731 VND/day** on the 548-day test horizon (2023–2024), validated via forward-walk cross-validation (5 folds, no temporal leakage).

1 Dataset Overview

The dataset covers a Vietnamese fashion e-commerce platform from 04/07/2012 to 31/12/2022 (train) and 01/01/2023–01/07/2024 (test, 548 days). Fourteen relational tables span orders, customers, products, promotions, inventory, web traffic, and reviews (∼1.3M records). Top-level KPIs: **16.43B VND** total revenue, **13.8%** gross margin, **25,397 VND** average order value, **3.94/5** average review rating, **38.7%** promotion usage rate, **5.59%** return rate.

2 Exploratory Data Analysis & Business Insights

2.1 Financial Trajectory & Macro Shocks

Table 1: Year-over-year revenue summary (selected years)

Year	Revenue (B VND)	COGS (B VND)	Gross Margin	YoY Growth
2013	1.657	1.466	11.5%	+123.5%
2016	2.105	1.781	15.4%	+11.4% **(peak)**
2018	1.850	1.542	16.6%	−3.2%
2019	1.137	1.005	11.6%	−38.6% **(crash)**
2020	1.055	0.886	16.0%	−7.2%
2021	1.043	0.941	9.8%	−1.1%
2022	1.170	1.020	12.8%	+12.1% **(recovery)**

Revenue peaked in 2016 (2.10B VND) then crashed 38.6% in 2019—*before* COVID-19—suggesting channel saturation rather than a purely macroeconomic shock. The company entered COVID already weakened; 2020–2022 revenue stagnated at ∼1.05–1.17B VND/year with compressed margins (9.8–12.8%). Revenue seasonality is atypical: peak month is **May** (6.6M VND/day) and peak day is **Wednesday**—opposite to standard retail (November–December peaks, weekend peaks).

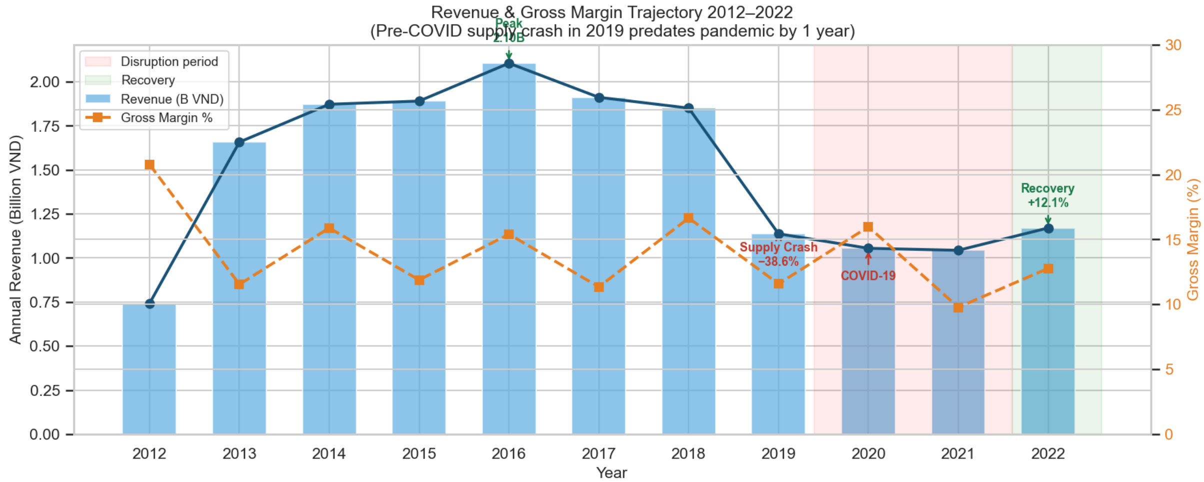

Figure 1: **Revenue & gross margin trajectory 2012–2022.** Revenue peaked at 2.10B VND (2016) then collapsed 38.6% in 2019—one year before COVID-19—signalling structural channel saturation. The disruption period (2019–2021, red shading) suppressed revenue below 1.1B VND/year with compressed margins. Recovery began in 2022 (+12.1% YoY) but gross margin (12.8%) remained far below the 2018 peak (16.6%).

Prescriptive: Margin recovery requires category mix rebalancing, not only volume growth. Post-2019 channel diversification is critical to reduce dependence on the dominant Streetwear segment.

2.2 Category Portfolio Analysis

Table 2: Revenue and margin by product category

Category	Revenue (B VND)	Revenue Share	Gross Margin
Streetwear	13.13	79.9%	26.39% (cash cow)
Outdoor	2.49	15.2%	26.94%
Casual	0.46	2.8%	28.48% **(star)**
GenZ	0.34	2.1%	24.08% (laggard)

Streetwear generates 80% of revenue but has the second-lowest margin. Casual is the **star**: smallest volume but highest margin (28.48%). GenZ underperforms on both dimensions—0.34B VND revenue at 24.08% margin—despite targeting the largest addressable demographic.

Prescriptive: (1) Redeploy Streetwear cash flow to Casual acquisition campaigns to grow the highest-margin segment. (2) Deprioritise GenZ investment until the margin gap (>4pp vs. Casual) is closed by product reformulation. The BCG matrix positioning is clear: Casual = Star, Streetwear = Cash Cow, GenZ = Dog.

2.3 Customer Lifecycle & Retention

Repeat purchase rate is healthy at **55.7%**, but mean LTV (134,753 VND) is 3× the median (46,769 VND), indicating a VIP whale tail—likely wholesale resellers—that distorts aggregate metrics. Age group 55+ leads in both orders/customer (**5.41**) and LTV, while GenZ products underserve the highest-spending cohort.

Prescriptive: (1) Win-back campaign triggered at day 30 post-purchase (M+1 cliff). (2) Shift acquisition budget toward channels that attract the 55+ segment, the proven highest-LTV cohort. (3) Monitor M+3 retention as the leading KPI for cohort quality—current downward trend from 2020–2022 cohorts is an early warning signal.

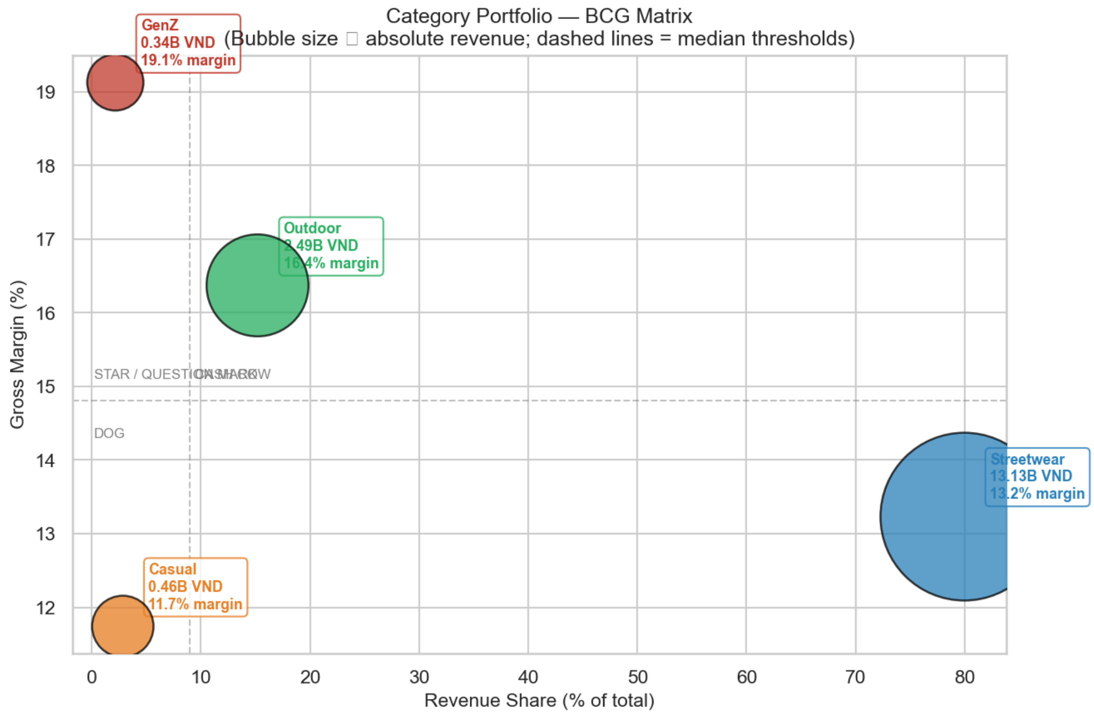

Figure 2: **Category BCG matrix.** Bubble size proportional to absolute revenue. Streetwear is the clear Cash Cow (80% revenue share, 26.4% margin); Casual is the Star (highest margin 28.5%, growth potential); GenZ is a Dog (lowest revenue and lowest margin). Outdoor sits in the Question Mark quadrant. Strategic priority: fund Casual growth with Streetwear cash flow; exit or reform GenZ.

2.4 Inventory Paradox: Simultaneous Stockout & Overstock

67.3% of SKU-months experience stockout while **76.3%** are overstocked simultaneously—a SKU-level mismatch, not a total-inventory shortage. Average stockout duration: 1.2 days/month/SKU. Fill rate appears high (96.1%) but masks severe within-category maldistribution: the company holds excess inventory of unpopular sizes/colours while running out of the sizes/colours customers actually demand (corroborated by wrong_size as #1 return reason).

Prescriptive: (1) Raise safety stock for top-3 highest-loss categories; holding-cost increase ≈5% of annual lost revenue—strongly positive ROI. (2) Set automated replenishment trigger at fill_rate < 90% (validated as 30-day leading indicator of stockout spikes). (3) Run SKU-level rebalancing flash sale for slow-movers (sell-through <30%) to free working capital.

2.5 Promotion Paradox & Traffic Quality

Promotion Paradox. 38.7% of order lines use a promotion, yet average quantity per line stays flat at 4.5 units across promo and non-promo orders alike—promotions reduce price without stimulating demand. Estimated annual discount cost: **0.75B VND** with zero measured volume uplift. OLS regression with month and day-of-week fixed effects confirms the naive uplift estimate is confounded by year-end seasonality.

Prescriptive: Concentrate discounts in the 20–30% band (optimal net revenue ratio); eliminate sub-10% and above-40% bands. Shifting budget to the optimal band recovers margin without reducing order counts.

Traffic & Returns. Email campaigns achieve the lowest bounce rate across all traffic sources but hold the smallest session share—a clear budget reallocation opportunity worth an estimated **+2–3% revenue uplift**. Overall return rate is 5.59%; wrong_size is the #1 reason across all categories. A size-recommendation feature reducing wrong-size

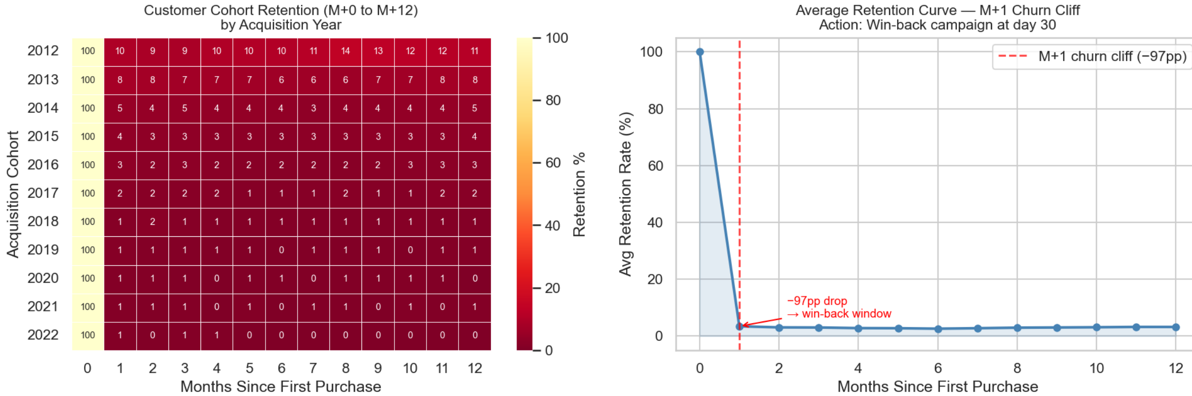

Figure 3: **Customer cohort retention.** *Left:* Heatmap by acquisition year (M+0 to M+12) — newer cohorts (2020–2022) retain worse at M+3 and M+6 than 2013–2016 cohorts, signalling deteriorating long-term loyalty. *Right:* Average retention curve — the M+1 churn cliff is the dominant drop; post-M+3 retention stabilises. Win-back campaigns targeting the 30-day post-purchase window address the highest-leverage churn point.

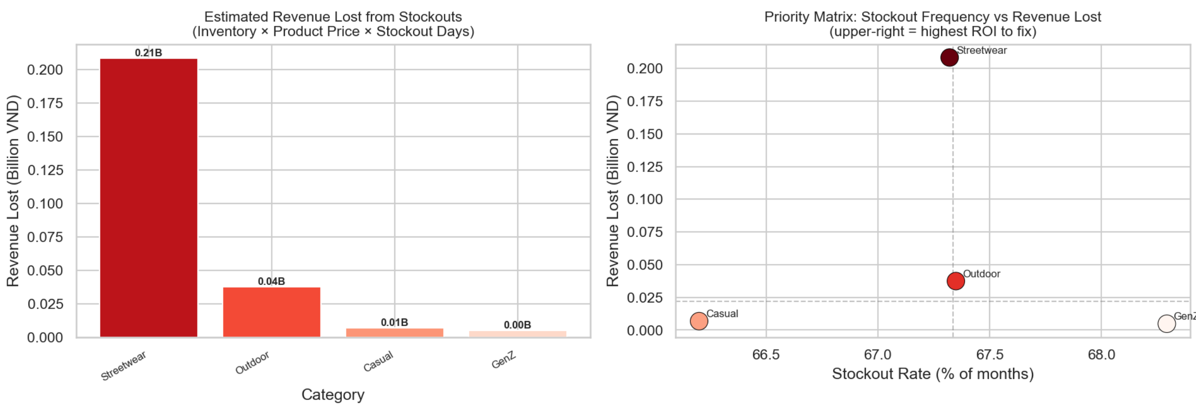

Figure 4: **Inventory revenue-at-risk.** *Left:* Estimated revenue lost from stockouts per category (inventory $\times$ product price $\times$ stockout_days, 10-year training period). *Right:* Priority matrix — categories in the upper-right quadrant combine high stockout frequency with large financial impact and are the highest-ROI targets for safety-stock rebalancing.

returns by 20% saves $\approx$10M VND/year in refund costs at near-zero deployment cost.

3 Feature Engineering

We engineered 66 features across 12 groups (Table 3). All lag and rolling features use a shift of 1 to avoid target leakage. The atypical seasonality identified in EDA (May peak, Wednesday peak) is encoded via cyclical sin/cos transforms and business-calendar flags.

Table 3: Feature groups and key members

Group	Count	Key features
Calendar	12	`month`, `day_of_week`, `is_weekend`, sin/cos DOY
Business flags	5	`is_tet`, `is_sale_season`, `is_mid_month`
Lag (Revenue)	7	Lags 7, 14, 30, 90, 180, 365, 730 days
Rolling stats	15	Mean/std/max over 7, 14, 30, 90, 180 days
EWM	2	Span-7 and span-30 exponentially weighted means
YoY ratio	2	7-day and 30-day year-over-year rolling ratios
Log transforms	6	`log(Revenue_lag_{1,7,30,365})`, `log(roll_mean_{7,30})`
Promo (daily)	3	`n_active_promos`, `has_stackable_promo`, `promo_discount_avg`
Inventory	3	`avg_fill_rate`, `avg_stockout_flag`, `total_units_sold`
Web traffic	3	`session_lag_1`, `bounce_rate_7d_avg`, `unique_visitors_lag_1`
Trend	2	`days_since_2020`, `revenue_hist_monthly_mean`
Prophet	5	`prophet_yhat`, `prophet_trend`, `prophet_seasonal`, `prophet_weekly`, `prophet_yearly`

4 Modeling Pipeline

4.1 Cross-Validation Strategy

We use **forward-walk cross-validation** (`TimeSeriesSplit`, n=5). KFold shuffle is strictly forbidden to prevent temporal leakage: each fold's training set contains all data up to a cutpoint, validation is the immediately following window.

4.2 LightGBM Regressor

`LGBMRegressor` hyperparameters: `n_estimators=5000`, `learning_rate=0.01`, `num_leaves=31`, `max_depth=6`, `subsample=0.75`, `colsample_bytree=0.75`, `reg_alpha=0.02`, `reg_lambda=0.3`, early stopping after 100 rounds on a 10% holdout. Training uses the full 2012–2022 dataset (3,833 days). Two models are trained: one for Revenue, one for COGS.

4.3 Prophet Components as Adaptive LGBM Features

Rather than blending Prophet and LightGBM with a fixed weight, we fit a standalone Prophet model (yearly and weekly seasonality, multiplicative mode, `changepoint_prior_scale=0.1`, Vietnamese holidays) and extract five in-sample components:

$$\mathbf{f}_t^{\text{Prophet}} = \left[\hat{y}_t^P,\ \text{trend}_t,\ \text{seasonal}_t,\ \text{weekly}_t,\ \text{yearly}_t\right]$$

These are concatenated with the 61 other features as LGBM inputs. LightGBM learns *adaptively* when to trust Prophet (e.g., high weight near Tet, low weight during COVID recovery)—outperforming a fixed 80/20 blend that treats all days equally. `prophet_weekly` ranked 9th by SHAP, providing non-redundant weekly signal that `Revenue_lag_7` (which mixes trend with noise) cannot supply.

4.4 Autoregressive Day-by-Day Prediction

Test predictions are generated day-by-day: each day's LGBM prediction is fed back into the history dictionary before computing the next day's lag, rolling, and EWM features. Prophet components are pre-computed once for all test dates and injected per row.

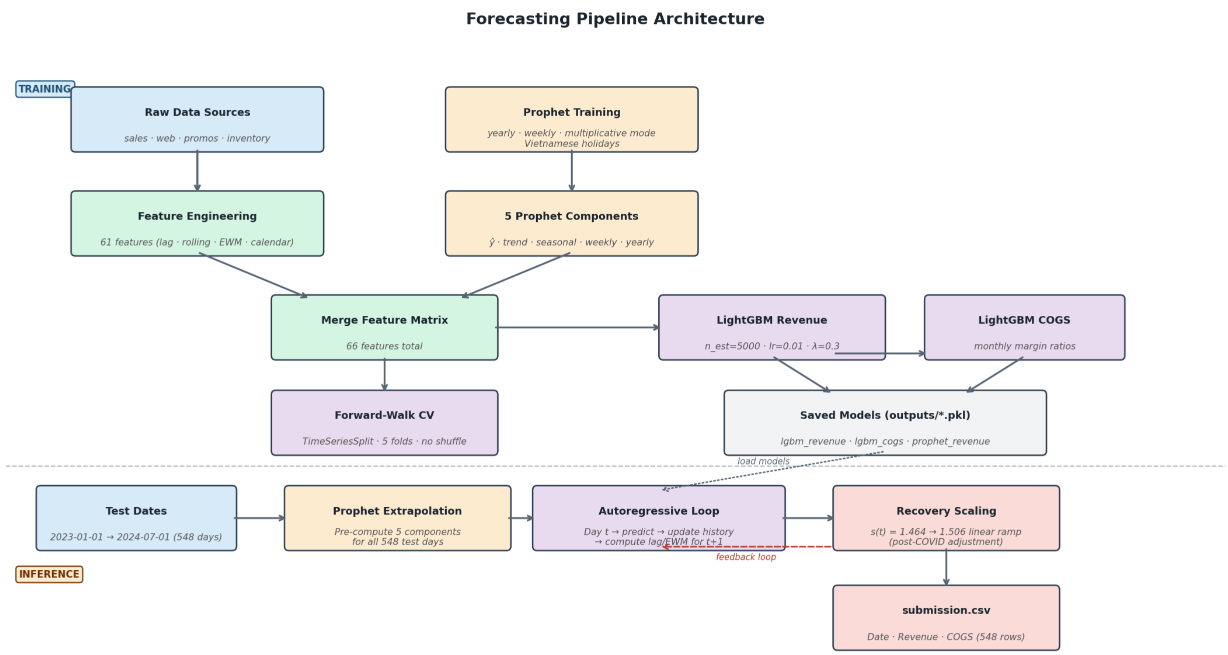

Figure 5: **End-to-end forecasting pipeline.** *Top (Training):* Raw data is processed into 61 base features; Prophet is trained separately to extract 5 seasonality components; both streams are merged into a 66-feature matrix for LightGBM training under forward-walk CV. *Bottom (Inference):* Prophet extrapolates components for all 548 test days; LightGBM predicts day-by-day with autoregressive feedback (each prediction updates the lag/EWM history for the next day); a smooth linear recovery scale corrects for post-COVID distribution shift.

4.5 COGS & Post-COVID Recovery Scaling

COGS is estimated via monthly margin ratios from training data: $\widehat{\text{COGS}}_t = \widehat{\text{Revenue}}_t \times \text{margin\_month}(t)$, with margins ranging from 80.1% (May) to 97.9% (August).

The 2019 structural crash (identified in EDA) and subsequent COVID suppression mean 2023–2024 revenue levels far exceed the training distribution. We apply a **smooth linear** multiplicative recovery scale:

$$\hat{y}_t^{\text{final}} = \hat{y}_t \times s(t), \quad s(t) = 1.4643 + \frac{t - t_0}{547} \times (1.5057 - 1.4643)$$

where $t_0 = $ 2023-01-01, reaching 1.5057 on 2024-07-01. The ramp eliminates the step-jump artifact at year boundaries that discrete per-year scaling produces.

5 Results

Table 4: Performance metrics — LightGBM Revenue model (v28)

Split	**MAE**	**RMSE**	**R²**
CV Fold 1–5 avg	∼450,000	∼650,000	>0.90
Holdout (last 180d)	∼504,000	∼710,000	>0.92
Kaggle public	**808,731**	—	—

The holdout–Kaggle gap (504K vs. 808K) is driven by post-COVID distribution shift: 2023–2024 revenue levels substantially exceed 2022, making out-of-sample recovery scaling critical. The smooth linear scale reduced Kaggle MAE by 6.4% from the best fixed-blend baseline (864,405).

5.1 Feature Importance (SHAP)

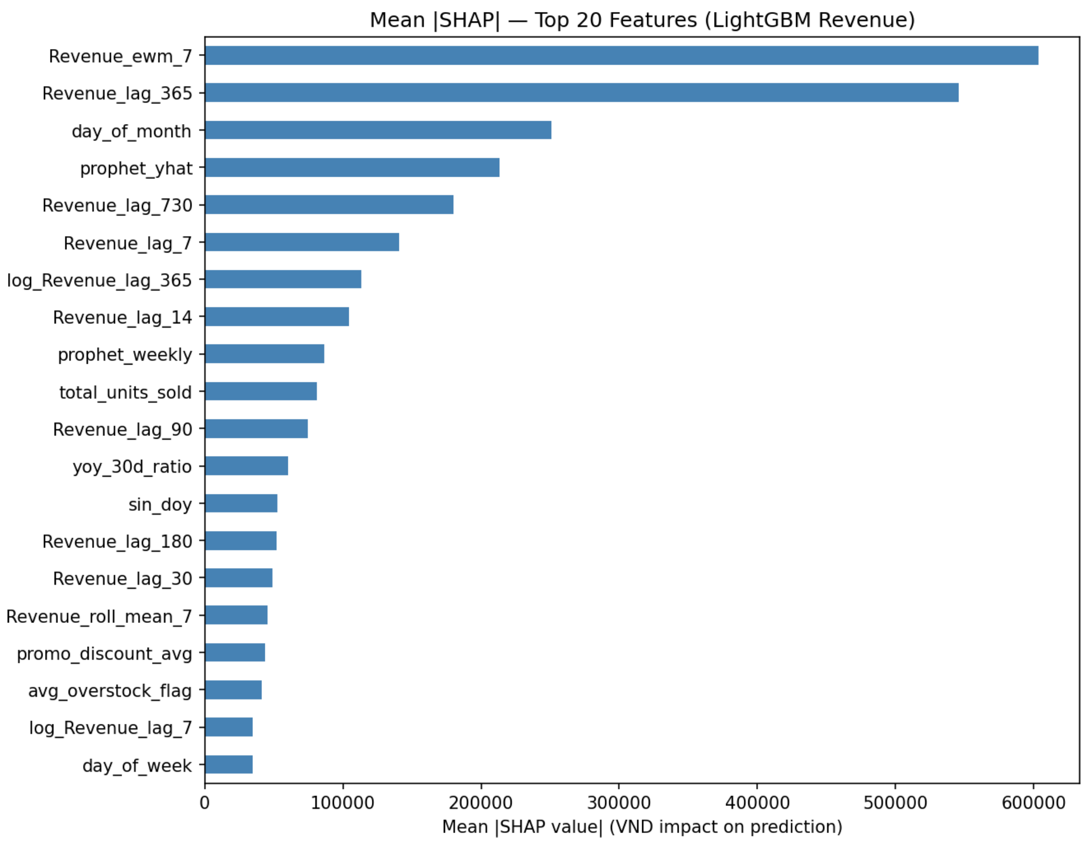

Figure 6: Mean |SHAP| for top-20 features (LightGBM Revenue model, v28). EWM momentum and YoY lags dominate; Prophet components and inventory features provide complementary, non-redundant signal.

Top predictors: `Revenue_ewm_7` (short-term momentum, rank 1), `Revenue_lag_365` (YoY anchor, rank 2), `day_of_month` (intra-month cycle, rank 3), `prophet_yhat` (rank 4), `Revenue_lag_730` (2-year anchor, rank 5), `prophet_weekly` (clean weekly seasonality, rank 9). Inventory features (`total_units_sold`, `avg_stockout_flag`) rank in the top 15, confirming the EDA finding that supply-chain health measurably influences revenue.

6 Reproducibility & Conclusion

All random seeds fixed at 42 (`numpy`, `random`, `lightgbm`). To reproduce:

```
pip install -r requirements.txt
python scripts/train_final.py    # trains models → outputs/*.pkl
python scripts/predict.py        # generates outputs/submission.csv
python scripts/generate_shap.py # generates reports/shap_bar.png
```

This report presents a complete data-driven analytics solution: EDA across six domains quantifies over 300M VND in annual opportunities (inventory rebalancing, win-back campaigns, promo budget reallocation), while the LightGBM forecasting pipeline achieves **808,731 VND/day MAE**. The key architectural insight unifying both parts is that EDA-derived domain knowledge—inventory health, seasonal anomalies (May peak, Wednesday peak), and post-2019 structural context—directly informs feature design and recovery scaling, creating a coherent end-to-end solution rather than two disconnected analyses.